



Back-End AI Engineering Internship

FlyRank AI · IILM University, Greater Noida

Maulik Pandey · Roll No. 25SCS1003003495 · B.Tech CSE (AI/ML), Batch 2025 - 2029

1 July - 26 August 2026 · 8 Weeks

About the Intern

Name

Maulik Pandey

Roll Number

25SCS1003003495

Institute

IILM University, Greater Noida, U.P.

Programme

B.Tech CSE (AI/ML), Batch 2025 - 2029

Role

Back-End AI Engineering Intern

Duration

8 Weeks · 1 July - 26 August 2026

COMPANY OVERVIEW

FlyRank AI

FlyRank is an AI-native organic-growth platform that coordinates specialized AI agents across strategy, content production, optimization, indexation, and continuous improvement. Its technology focus makes backend reliability, data flows, and usage infrastructure important to AI-enabled product systems.



AI-Native Workflows

Coordinates specialized agents around organic growth and search visibility.



Search & Growth

Supports D2C and B2B brands with AI-assisted organic-growth workflows.



Engineering Focus

Backend services provide reliable APIs, data persistence, metering and integrations.

Official website: flyrank.ai

Reliable Usage Metering & Billing for AI Services

AI-enabled products may need to bill customers according to variable API and token usage.



The four-step pipeline ensures accurate tracking, plan enforcement, deterministic billing, and safe handling of edge cases like retries and duplicate requests.

Multi-Tenant Usage Metering & Billing Engine

Multi-Tenancy

Keep usage and subscription state isolated by tenant.

Usage Metering

Record API calls and AI-token usage.

Quota Enforcement

Apply plan-specific monthly usage limits.

Exact Billing

Calculate charges with integer-cent arithmetic.

Idempotency

Prevent duplicate billable requests and conflicts.

Stripe Integration

Support Checkout and secure subscription webhooks.

The Toolkit Behind the Internship



TypeScript

Type-safe backend development



Node.js + Express

REST API and server runtime



PostgreSQL + pg

Relational persistence and transactions



Knex.js

Database migrations and seed data



Stripe

Checkout, subscriptions and webhooks



Zod

Runtime request/data validation



Vitest + Supertest

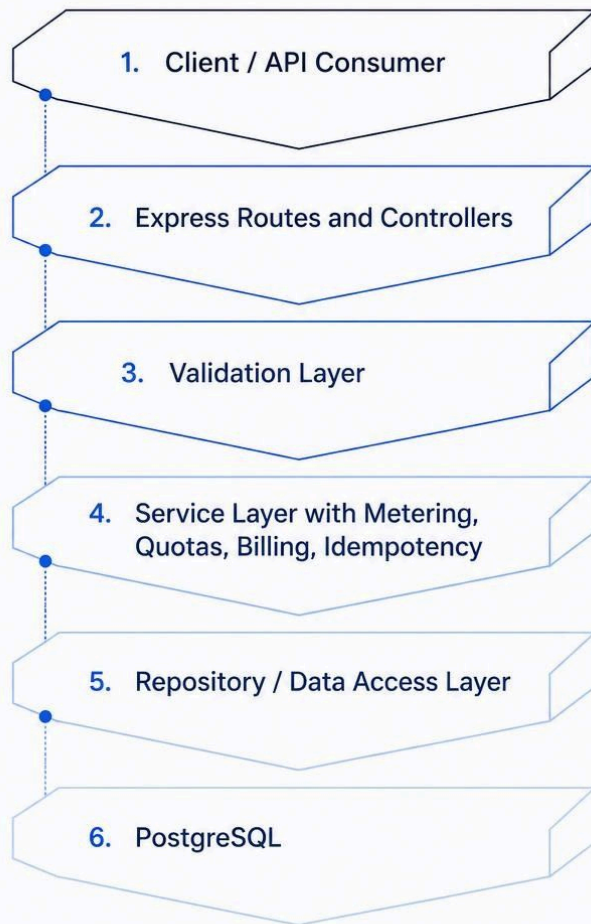
Unit and API/integration testing



Docker + GitHub

Reproducible environment and version control

From API Request to Metered Billing



System Architecture

A six-layer architecture processes every API request from the client through validation, service logic (metering, quotas, billing, idempotency), and data access, finally persisting to PostgreSQL.

- ❏ **Core Principle:** Keep Stripe payment integration isolated from core metering logic – handling Checkout sessions, subscription state, signed webhooks, and replay protection independently.

The 8-Week Engineering Journey

Week 1 — Onboarding & Architecture

Project context, backend structure, requirements

1

2

Week 2 — Database Foundations

PostgreSQL schema, migrations, repositories

3

Week 3 — Usage Metering

API calls, AI-token usage, tenant context

4

Week 4 — Quotas & Billing

Plans, limits, deterministic pricing

5

Week 5 — Reliability

Idempotency, fingerprints, transactions

6

Week 6 — Stripe Integration

Checkout, subscriptions, signed webhooks

7

Week 7 — Testing & Reconciliation

Unit/integration tests, edge cases

8

Week 8 — Capstone Completion

End-to-end validation, docs and final cleanup

OUTCOMES

Key Engineering Work & Outcomes

Implemented

- Tenant-aware API and usage-event processing
- Monthly quota enforcement by subscription plan
- Idempotency keys and request fingerprinting
- Integer-cent monetary calculations
- Stripe Checkout and subscription synchronization
- Webhook signature verification and replay protection

Engineering Outcomes

- PostgreSQL persistence with transactional workflows
- Automated unit and integration tests
- Validation with Zod
- Docker-based development environment
- Git/GitHub-based version-controlled workflow
- Documented backend structure and reproducible setup

COMPLETION

Key Learnings & Completion

Key Learnings

- Backend architecture and API design
- PostgreSQL and transactional processing
- Usage metering and subscription billing
- Idempotency and reliable retries
- Secure third-party webhooks
- Automated testing and debugging

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1 July 2026 - 26 August 2026

Thank You

Back-End AI Engineering Internship FlyRank AI

Maulik Pandey

25SCS1003003495 | IILM University

Capstone: Multi-Tenant Usage Metering and Billing Engine

Internship Documentation

Offer Letter

FlyRank^{AI}

June 26, 2026

Internship confirmation

Re: Internship Confirmation - Maulik Pandey, July 2026

To whom it may concern,

This letter confirms that **Maulik Pandey** (maulikpandey2007@proton.me) has been accepted into the FlyRank AI Internship program as a **Back-End AI Engineering Intern** for July 2026.

Intern name	Maulik Pandey
Role / title	Back-End AI Engineering Intern
Program	July 2026 - Backend AI Engineering
Start date	July 1, 2026
End date	August 26, 2026
Duration	8 weeks
Status	Approved

Participation confirms acceptance into the internship program and does not create employment, contractor status, wages, benefits, or a job guarantee.

Schools, universities, and employers verifying this confirmation can email verify@internship.flyrank.ai.



Alen Malkoc, Founder, CEO
FlyRank Corp.

Completion Certificate

FlyRank^{AI}

ID: FR-D11-E8473-2D769
Date Issued: 10.09.2026.

Certificate of Completion

This is to certify that

Maulik Pandey

has successfully completed the

Backend AI Engineering

Internship Program at FlyRank.ai from 01.07.2026. to 26.08.2026.

The intern has demonstrated excellence in technical competency, professional conduct, and collaborative contribution to the FlyRank team.



Alen Malkoc, Founder & CEO, FlyRank Corp.
FlyRank Corp.



FlyRank Corp. AI Internship Program www.flyrank.ai

Verify this certificate at <https://internship.flyrank.ai/verify> using ID FR-D11-E8473-2D769

Made with GAMMA

THANK YOU

Gratitude & Reflections

My sincere thanks to the entire team at **FlyRank AI** for an invaluable Back-End AI Engineering Internship experience.

I am deeply grateful for the mentorship, guidance, and the opportunity to contribute to a challenging and innovative project. This internship has significantly enhanced my skills in backend development, AI service architecture, and reliable system design.

With appreciation,

Maulik Pandey

FlyRank AI Intern

1 July - 26 August 2026